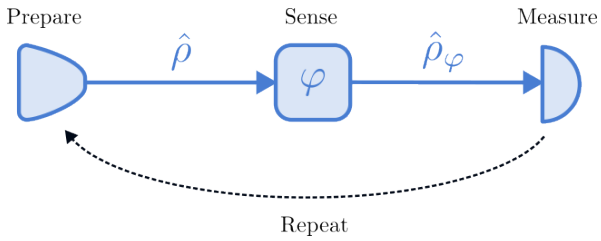

Figure 4.3: The main steps of a quantum sensing protocol. First, a probe system is prepared in some initial state $\hat{\rho}$. Then, the probe undergoes parameter-dependent evolution resulting in a parameter-dependent state $\hat{\rho}_\varphi$. Finally, the probe is measured. The whole process is repeated a number of times to gather data, from which the parameter value is estimated.

probability distribution over possible parameter values which is then repeatedly updated based on the observed data.

4.1.5 Quantum Fisher information

In quantum sensing, the parameter of interest φ will be picked up by the state $\hat{\rho}_\varphi$ of some probe system, as illustrated in Fig. 4.3, and the probe is then measured. Each particular measurement that may be performed on the probe corresponds to some probability distribution, and the amount of information about the parameter in this distribution can be quantified by the Fisher information. However, there are (infinitely) many possible measurements. Some will be better at extracting information about φ than others. It is therefore natural to ask how much information about φ is contained in the quantum state $\hat{\rho}_\varphi$, rather than in a particular outcome distribution. This is quantified by the quantum Fisher information.

The *quantum Fisher information* (QFI) of a parametrised family of states $\hat{\rho}_\varphi$ with respect to the parameter φ is the maximum classical Fisher information over all possible measurements

$$\mathcal{Q}_\varphi(\hat{\rho}_\varphi) = \max_{\{\hat{E}_x\}} \mathcal{F}_\varphi(p_\varphi) \tag{4.34}$$

subject to $\{\hat{E}_x\}$ being a POVM and $p_\varphi(x) = \mathrm{Tr}[\hat{\rho}_\varphi \hat{E}_x]$. In other words, $\mathcal{Q}_\varphi$ is the classical Fisher information corresponding to the optimal measurement. The *quantum Cramér-Rao bound* states that, for any unbiased estimator and any measurement on $\hat{\rho}_\varphi$,

$$\langle \Delta^2 \tilde{\varphi} \rangle \geq \frac{1}{\mathcal{Q}_\varphi(\hat{\rho}_\varphi)}. \tag{4.35}$$

This follows directly from the classical Cramér-Rao bound.

Computing the QFI

A natural question is whether the optimisation in the definition (4.34) of the QFI can be performed in general. A priori, this looks like a difficult problem since the dimension of the system may be large and parametrising all possible

POVMs will introduce many optimisation variables. Surprisingly, as shown by Braunstein and Caves [2], the QFI can be expressed in closed form.

In general, the QFI can be written

$$\mathcal{Q}_\varphi(\hat{\rho}_\varphi) = \mathrm{Tr}[\hat{\rho}_\varphi \hat{L}^2_\varphi(\hat{\rho}_\varphi)], \tag{4.36}$$

where $\hat{L}_\varphi(\hat{\rho}_\varphi)$ is the so-called *symmetric logarithmic derivative*, which is a Hermitian operator implicitly defined by $\dot{\hat{\rho}}_\varphi = \frac{1}{2}\left[\hat{\rho}_\varphi \hat{L}_\varphi(\hat{\rho}_\varphi) + \hat{L}_\varphi(\hat{\rho}_\varphi)\hat{\rho}_\varphi\right]$, where the dot denotes derivative with respect to φ. It can be expressed explicitly in the eigenbasis of $\hat{\rho}_\varphi$ as

$$\hat{L}_\varphi(\hat{\rho}_\varphi) = 2\sum_{n,m} \frac{\langle \lambda_{\varphi,m}|\dot{\hat{\rho}}_\varphi|\lambda_{\varphi,n}\rangle}{\lambda_{\varphi,m}+\lambda_{\varphi,n}} |\lambda_{m,\varphi}\rangle\langle\lambda_{n,\varphi}|, \tag{4.37}$$

where $\lambda_{\varphi,m}$ and $|\lambda_{\varphi,m}\rangle$ are the eigenvalues and corresponding eigenstates of $\hat{\rho}_\varphi$, and the sum is only over terms for which $\lambda_{\varphi,m}+\lambda_{\varphi,n}$ is non-zero. The QFI can then be expressed

$$\mathcal{Q}_\varphi(\hat{\rho}_\varphi) = 2\sum_{n,m} \frac{|\langle \lambda_{\varphi,m}|\dot{\hat{\rho}}_\varphi|\lambda_{\varphi,n}\rangle|^2}{\lambda_{\varphi,m}+\lambda_{\varphi,n}} \tag{4.38}$$

$$= \sum_n \frac{\dot{\lambda}^2_{\varphi,m}}{\lambda_{\varphi,m}} + 2\sum_{n,m} \frac{(\lambda_{\varphi,m}-\lambda_{\varphi,n})^2}{\lambda_{\varphi,m}+\lambda_{\varphi,n}} |\langle \dot{\lambda}_{\varphi,m}|\lambda_{\varphi,n}\rangle|^2, \tag{4.39}$$

where again the sums are only over terms with non-zero denominator. In the last line, the first term can be understood as the classical Fisher information associated with the probability distribution over eigenvalues coming from the mixedness of $\hat{\rho}_\varphi$, while the second term is a 'genuinely quantum' contribution. For pure states $\hat{\rho}_\varphi = |\psi_\varphi\rangle\langle\psi_\varphi|$, the symmetric logarithmic derivative becomes

$$\hat{L}_\varphi(\hat{\rho}_\varphi) = 2\left(|\dot{\psi_\varphi}\rangle\langle\psi_\varphi| + |\psi_\varphi\rangle\langle\dot{\psi_\varphi}|\right) \tag{4.40}$$

and the QFI simplifies to

$$\mathcal{Q}_\varphi(\hat{\rho}_\varphi) = 4\left(\langle\dot{\psi_\varphi}|\dot{\psi_\varphi}\rangle - |\langle\dot{\psi_\varphi}|\psi_\varphi\rangle|^2\right). \tag{4.41}$$

Unitary encoding

An important class of estimation problems are those where the parameter is encoded onto the probe via unitary evolution. That is, the parameter is imparted onto some initial state by a unitary $\hat{U}_\varphi = e^{-i\varphi\hat{H}}$ for some Hermitian operator $\hat{H}$. If input is a pure state $|\psi\rangle$, the output state is also pure

$$|\psi_\varphi\rangle = \hat{U}_\varphi|\psi\rangle = e^{-i\varphi\hat{H}}|\psi\rangle. \tag{4.42}$$

In this case, for finite-dimensional systems, it can be shown (see Prob. 4.5) that the QFI is given by

$$\mathcal{Q}_\varphi = 4\left[\langle\psi|\hat{H}^2|\psi\rangle - \langle\psi|\hat{H}|\psi\rangle^2\right] = 4\langle\Delta^2\hat{H}\rangle_\psi. \tag{4.43}$$

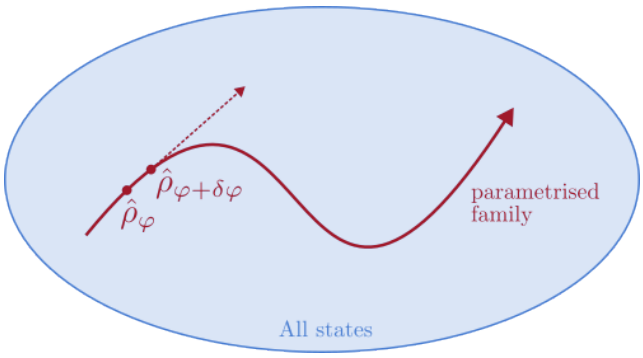

Figure 4.4: A parametrised family of states defines a trajectory within the space of all states (of given dimension) The QFI $\mathcal{Q}_\varphi(\hat{\rho}_\varphi)$ can be understood as quantifying the rate of change at each point along this trajectory.

The optimal input state, maximising the QFI, is

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|\lambda_{\min}\rangle + |\lambda_{\max}\rangle), \tag{4.44}$$

where $\lambda_{\min}$ and $\lambda_{\max}$ are the smallest and largest eigenvalues of $\hat{H}$, respectively. The optimal QFI is $(\lambda_{\max} - \lambda_{\min})^2$.

Additivity and convexity

Just as the clasical Fisher information is additive for independent random variables, the QFI is additive for product states

$$\mathcal{Q}_\varphi(\hat{\rho}_\varphi \otimes \hat{\rho}'_\varphi) = \mathcal{Q}_\varphi(\hat{\rho}_\varphi) + \mathcal{Q}_\varphi(\hat{\rho}'_\varphi). \tag{4.45}$$

The QFI is also convex under parameter-independent mixing

$$\mathcal{Q}_\varphi(\sum_k q_k \hat{\rho}_\varphi^{(k)}) \leq \sum_k q_k \mathcal{Q}_\varphi(\hat{\rho}_\varphi^{(k)}) \tag{4.46}$$

where the q_k define a probability distribution, $\sum_k q_k = 1$, $q_k \geq 0$, and do not depend on φ. This convexity property implies that for encodings that do not introduce parameter-dependent mixing – in particular unitary encodings – pure input states are always optimal.

Relation to Bures angle and speed of change

Intuitively, the more a state $\hat{\rho}_\varphi$ changes with changes in the parameter φ, the larger the QFI should be. This is indeed the case. The QFI can be understood as quantifying the speed of change along the trajectory of $\hat{\rho}_\varphi$ in state space when varying φ, as illustrated in Fig. 4.4. This is seen by relating the QFI to a measure of closeness between density operators.

The *Bures angle* between two quantum states $\hat{\rho}$ and $\hat{\sigma}$ is given by $a_B(\hat{\rho}, \hat{\sigma}) = \arccos\sqrt{F(\hat{\rho}, \hat{\sigma})}$, where F is the fidelity as defined in Eq. (2.87). It generalises a notion of closeness between classical probability distributions (the

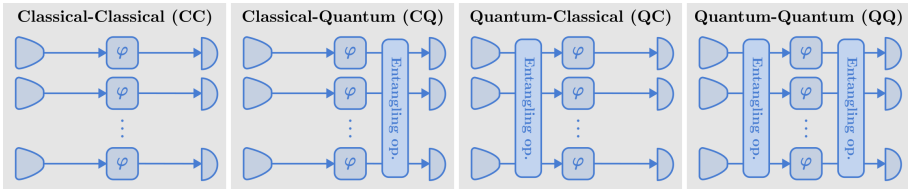

Figure 4.5: Four scenarious for quantum sensing with multiple probes. The probes are either prepared and measured individually (CC), prepared individually but jointly measured (CQ), prepared in an entangled state and measured individually (QC), or both initially entangled and jointly measured (QQ).

Bhattacharyya distance). For a small change in parameter

$$a_B(\hat{\rho}_\varphi, \hat{\rho}_{\varphi+\delta\varphi}) = \frac{1}{2}\sqrt{\mathcal{Q}_\varphi(\hat{\rho}_\varphi)}\delta\varphi + O(\delta\varphi^2), \tag{4.47}$$

which means that $\mathcal{Q}_\varphi(\hat{\rho}_\varphi)$ can be understood as four times the squared 'speed', $\mathcal{Q}_\varphi = 4(da_B/d\varphi)^2$, along the trajectory of $\hat{\rho}_\varphi$.

4.2 Precision scaling

In general, one is interested in obtaining the best possible estimate with the resources available. Resources may be e.g. the size or number of probe systems used, the number of repetitions of the experiment, or the total time abvailable to perform the experiment. It is therefore interesting to understand how the optimal estimate variance scales with these resources and, in particular, any difference in this scaling between classical and quantum approaches.

We can distinguish four general scenarios, as illustrated in Fig. 4.5. In each scenario, N probe systems are used and the experiment is repeated ν times. The N probes always individually undergo identical evolutions that depend on the parameter of interest φ. However, both the initial state preparation and the final measurement may be either individial or collective, giving four possibilities. In the context of precision scaling here, we refer to the individual strategies, which are not entangling, as 'classical' (C) and the collective, which can entangle systems, as 'quantum' (Q), labelling the scenarios CC, CQ, QC, and QQ, as shown[3]. How does the optimal estimate variance behave in each scenario?

In the CC and CQ scenarios, the N probes are prepared independently and evolve independently. The state after the parameter-dependent evolution is therefore a product state

$$\hat{\rho}_{out} = \hat{\rho}_\varphi \otimes \cdots \otimes \hat{\rho}_\varphi = \hat{\rho}_\varphi^{\otimes N}. \tag{4.48}$$

Due to the additivity property (4.45), the QFI of this state is

$$\mathcal{Q}_\varphi(\hat{\rho}_{out}) = \mathcal{Q}_\varphi(\hat{\rho}_\varphi^{\otimes N}) = N\mathcal{Q}_\varphi(\hat{\rho}_\varphi) = N\mathcal{Q}_1, \tag{4.49}$$

[3]Although sensing with single systems without entanglement can also be quantum in other senses, as discussed in other sections.